A MORPHISM FROM A SEPARATED ALGEBRAIC SPACE INTO A FORMAL ALGEBRAIC SPACE IS SEPARATED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof ends by listing the four general statements it used; that list lies inside the counted span and is reproduced verbatim, and each item is recorded as a standard prerequisite.

Editorial note. The counted spans carry 1 navigation stub beyond the listed items; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- A morphism from an algebraic space into a formal algebraic space is representable by algebraic spaces. Same chapter, Lemma lemma-space-to-formal-space, invoked at proof lines 3750-3752. Makes the assertion meaningful and allows base change to schemes.
- Stability of separatedness under base change and composition, monomorphisms are separated, and a morphism into the base is separated when the source is. Morphisms of Spaces, Lemmas spaces-morphisms-lemma-base-change-separated, spaces-morphisms-lemma-composition-separated, spaces-morphisms-lemma-monomorphism-separated and spaces-morphisms-lemma-separated-implies-morphism-separated, invoked at proof lines 3759-3763. Supply the reduction and the final step.

Source reference. Original lemma 0AIQ: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2727–2727. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03KL: sources/stacks-chow/tags.txt, source lines 4681–4681. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03KQ: sources/stacks-chow/tags.txt, source lines 4686–4686. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 042N: sources/stacks-chow/tags.txt, source lines 5334–5334. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 04ZI: sources/stacks-chow/tags.txt, source lines 6466–6466. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let X be a formal algebraic space over S. Let $U \to X$ be a morphism where U is a separated algebraic space over S. Then $U \to X$ is separated.*

Proof. The statement makes sense because $U \to X$ is representable by algebraic spaces (Lemma 0AIQ). Let T be a scheme and $T \to X$ a morphism. We have to show that $U \times_X T \to T$ is separated. Since $U \times_X T \to U \times_S T$ is a monomorphism, it suffices to show that $U \times_S T \to T$ is separated. As this is the base change of $U \to S$ this follows. We used in the argument above: Morphisms of Spaces, Lemmas 03KL, 03KQ, 042N, and 04ZI. □